

LESSON 11 / 第十一課

Decay Detection: From Assessment Level to Residual Geometry

腐朽偵測：從評估級別到剩餘幾何

Format. Short literature review. Claims that carry a number or a causal ranking are tied to a numbered source. Scale runs from the largest published pattern to the finest measured process. The lesson is not sequenced from the previous number and is not a regional case file unless the object itself is regional.

體例。短文獻綜述。帶數字或因果排序的陳述繫到編號文獻。尺度由最大已發表格局寫到最細已測過程。本課不由上一號遞推；除非對象本身是地域性的，否則不是單一地區檔案。

Abstract

摘要

Instruments do not return a verdict. They return a geometry that must still be read inside a target, a weather envelope and a time frame. Sonic tomography inverts travel times; Burcham and colleagues showed that geometry error in that inversion is not trivial. A resistograph is a line sample. Foam fill can fool a velocity map. t/R is a ratio, not a pass/fail law (Bond, Gruber; Huang and colleagues on tropical trunks). Koeser et al. found likelihood-of-failure ratings noisy across assessors even when advanced data are present. Residual wood is the object. The colour plot is an input.

儀器不返回判決。它們返回仍須在目標、天氣包絡與時間框架內被閱讀的幾何。聲波斷層反演走時；Burcham 及同事顯示該反演中的幾何誤差不是小事。阻抗鑽是一條線樣本。泡沫填充可以愚弄速度圖。 t/R 是比率，不是通過／失敗定律（Bond、Gruber；Huang 及同事論熱帶樹幹）。Koeser 等發現即使有進階數據，失效可能性評級在評估者之間仍嘈雜。剩餘木材才是對象。顏色圖是輸入。

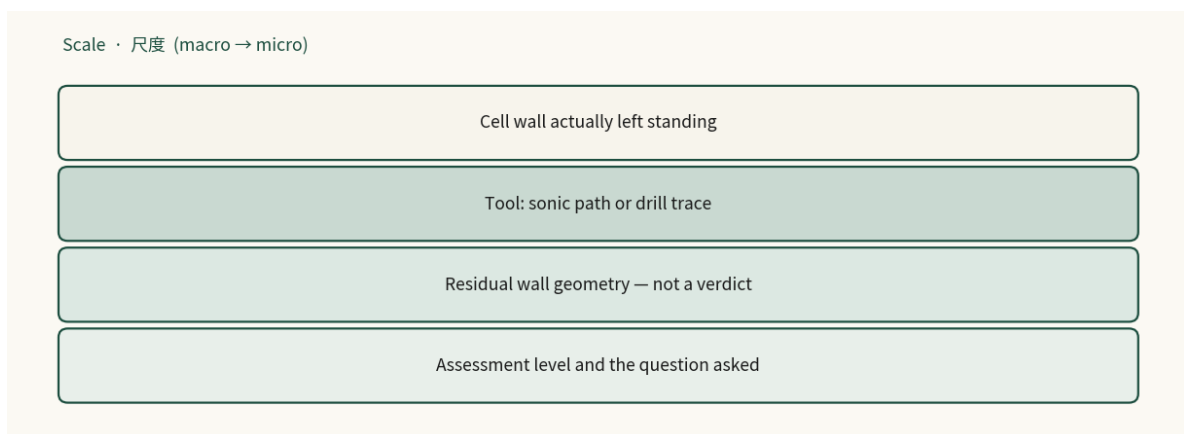


Figure 1. Nested scales from the question asked to residual cell wall.

圖 1. 由所問問題到剩餘細胞壁的嵌套尺度。

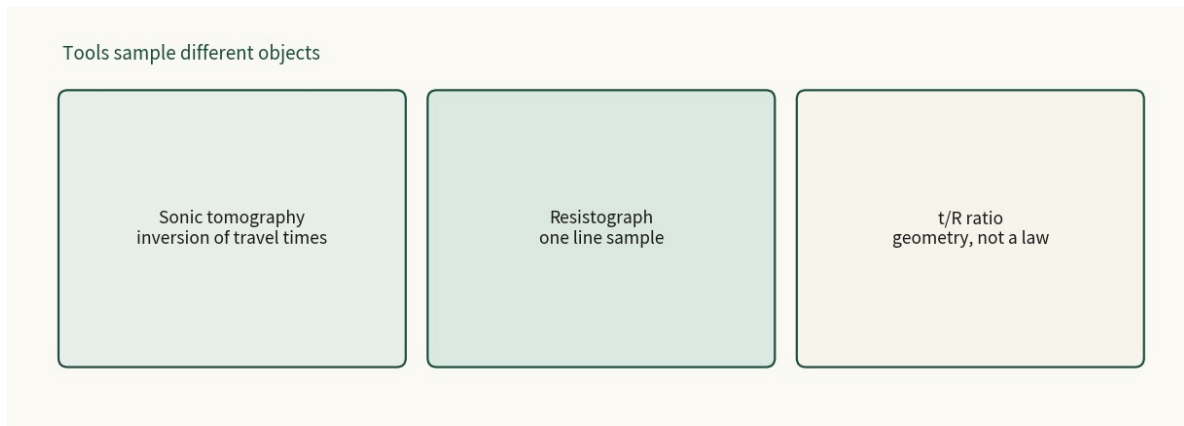


Figure 2. Tomography, resistograph and t/R sample different objects.

圖2. 斷層、阻抗鑽與 t/R 抽樣不同對象。

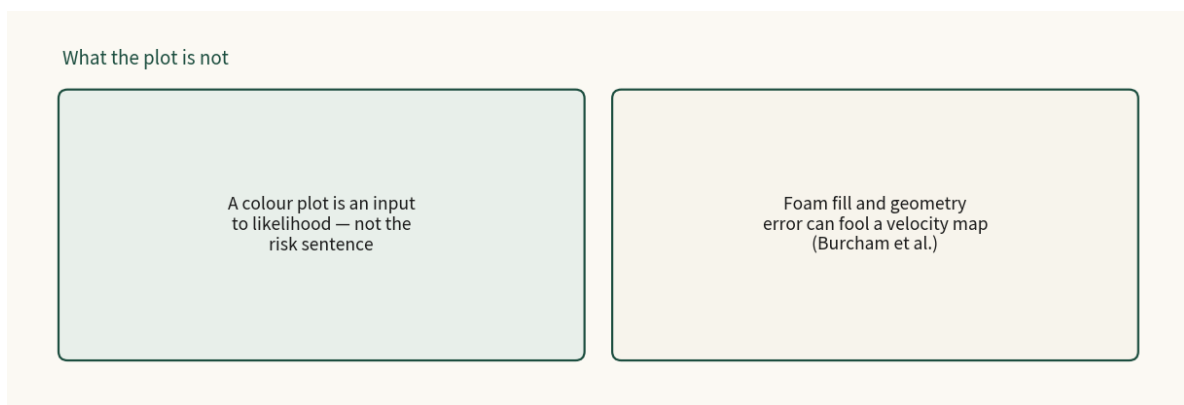


Figure 3. The plot informs likelihood; it does not close the sentence.

圖3. 圖為可能性提供資訊；它不結束句子。

1. Scope

1. 範圍

The object is published Level-3 decay detection and the residual geometry it claims to measure. A brand name is not a method.

對象是已發表的第3級腐朽偵測，以及它聲稱在測的剩餘幾何。品牌名不是方法。

2. What each tool sees

2. 各工具看見什麼

Sonic tomography inverts paths of mechanical waves; low-velocity zones are inferred, not photographed. Burcham et al. quantified geometry error in that inversion [1]. A resistograph records resistance along one borehole; it is a line sample whose interpretation depends on where the line was aimed [2]. Neither tool sees fungal identity. Neither tool sees the target field.

聲波斷層反演機械波路徑；低速區是推論出來的，不是拍到的。Burcham 等量化了該反演中的幾何誤差 [1]。阻抗鑽沿一條孔記錄阻力；它是線樣本，解釋取決於瞄準何處 [2]。兩種工具都看不見真菌身份。兩種工具都看不見目標場。

3. t/R and assessor noise

3. t/R 與評估者噪聲

Bond and Gruber argued that t/R thresholds treated as pass/fail laws are not supported as stand-alone criteria [3,4]. Huang and colleagues, working tropical trunks, likewise refused a single ratio as a verdict [5]. Koeser et al. measured disagreement among assessors on likelihood of failure; adding a plot does not automatically abolish that disagreement [6]. Residual load-bearing geometry still has to be named in the same sentence as the part, the weather and the target.

Bond 與 Gruber 主張把 t/R 閾值當成通過／失敗定律，並不作為獨立準則成立 [3,4]。Huang 及同事處理熱帶樹幹時，同樣拒絕單一比率作為判決 [5]。Koeser 等測量了評估者在失效可能性上的分歧；加上一張圖不會自動消滅該分歧 [6]。剩餘承載幾何仍須與部位、天氣與目標寫在同一句子裡。

4. Limits

4. 限度

Foam, cracks, irregular cross-sections and included bark all perturb velocity fields. A method that cannot declare those failure modes is not yet a method. Advanced data raise the resolution of the geometry account; they do not reprint capacity.

泡沫、裂縫、不規則截面與夾皮都會扰动速度場。不能宣布這些失效模式的方法，還不是方法。進階數據提高幾何賬戶的解析度；它們不重印容量。

Study questions

研習題

Choose the option that retains the published result. Answers follow the last item.

選出保留已發表結果的選項。答案在最後一題之後。

1. A tomogram in this review is:

1. 本綜述中的斷層圖是：

- A. A photograph of decay.
A. 腐朽照片。
- B. An inversion of travel times.
B. 走時反演。
- C. A legal verdict.
C. 法律判決。
- D. A P50.
D. 一個 P50。

2. Burcham et al. found geometry error in sonic tomography:

2. Burcham 等發現聲波斷層的幾何誤差：

- A. Trivial.
A. 微不足道。
- B. Not trivial.
B. 不是小事。
- C. Proof of zero decay.
C. 零腐朽的證明。
- D. A diversity rule.
D. 多樣性規則。

3. A resistograph trace is:

3. 阻抗鑽跡是：

A. A full 3-D map.

A. 完整三維圖。

B. A line sample.

B. 一條線樣本。

C. A target.

C. 一個目標。

D. Wall 4.

D. 第 4 壁。

4. t/R is treated as:

4. t/R 被視為：

A. A stand-alone pass/fail law.

A. 獨立通過／失敗定律。

B. Geometry that informs likelihood, not a law.

B. 為可能性提供資訊的幾何，不是定律。

C. An inoculant dose.

C. 接種劑量。

D. 3-30-300.

D. 3-30-300。

5. Koeser et al. found LoF ratings:

5. Koeser 等發現失效可能性評級：

A. Identical across assessors.

A. 評估者之間完全相同。

B. Noisy across assessors.

B. 在評估者之間嘈雜。

C. Unnecessary.

C. 不必要。

D. Equal to canopy %.

D. 等於樹冠%。

6. Foam fill can:

6. 泡沫填充可以：

A. Never affect a velocity map.

A. 永不影響速度圖。

B. Fool a velocity map.

B. 愚弄速度圖。

C. Raise P88.

C. 提高 P88。

D. Set SOS.

D. 設定 SOS。

7. Instruments see fungal identity:

7. 儀器看見真菌身份：

A. Always.

- A. 總是。
- B. Not by themselves.
- B. 本身看不見。
- C. When painted.
- C. 上漆時。
- D. In palms only.
- D. 只在棕櫚。

8. Advanced data:

8. 進階數據：

- A. Reprint capacity.
- A. 重印容量。
- B. Raise geometric resolution; they do not reprint capacity.
- B. 提高幾何解析度；不重印容量。
- C. Replace targets.
- C. 取代目標。
- D. End inspection.
- D. 結束檢查。

9. Residual wood is:

9. 剩餘木材是：

- A. A colour.
- A. 一種顏色。
- B. The object the review is after.
- B. 本綜述所追的對象。
- C. A catalogue.
- C. 一本目錄。
- D. A sacred month.
- D. 神聖月份。

10. A colour plot used as the whole assessment:

10. 把顏色圖當成整個評估：

- A. Completes the risk sentence.
- A. 完成風險句子。
- B. Skips target and weather envelope.
- B. 跳過目標與天氣包絡。
- C. Equals Choat 2012.
- C. 等於 Choat 2012。
- D. Is Wall 1.
- D. 是第 1 壁。

Answers / 答案 1-B 2-B 3-B 4-B 5-B 6-B 7-B 8-B 9-B 10-B

References

參考文獻

[1] Burcham D.C. et al. (2023). geometry error in sonic tomography.

[2] Rinn F. resistograph method papers.

[3] Bond J. (2006). t/R critique.

[4] Gruber F. (2008). t/R cannot predict stem failure.

[5] Huang Y.-S. et al. (2017). residual wall / tropical trunk geometry.

[6] Koeser A.K. et al. (2017). likelihood-of-failure rating studies.

[7] ANSI A300 Part 9 Level-3 language.

[8] Johnstone D., Moore G., Tausz M., Nicolas M. reviews of decay-detection tools.

Closing note. This lesson is written as a short review article. Disagreement in the literature is kept in view where the results actually diverge.

結語。本課以短綜述文章寫成。文獻真正分歧之處，兩造結果都保留在視野內。